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Graded Assignment • 30 min

Due Mar 18, 11:59 PM CET

1. Which action is better to cope with the so called "cocktail effect"? 1 point

- ○ Shutdown every transmission
- ◉ Scheduling of transmission (e.g., round-robin way of transmission, allowing only some group of transmitters to transmit, etc)

2. In a single channel radio network, as the number of communicating pairs increases, it is impossible to support every pair. Which is the better solution for such overwhelming interference scenario? 1 point

- ◉ Shrink the communication distance (such as multi-hop relay)
- ○ Make every node "move" randomly and wait for the source node to be close to its destination node.

3. Why does the cellular system look like as of today? 1 point

- ◉ To handle large amount of interference
- ○ To save the power consumption of mobile terminals

4. What are the drawbacks to have stationary access points in the wireless network? 1 point

- ○ Increasing transmission power of mobile terminals.
- ◉ Frequent handover (switching to a new access point) as mobile terminals move around the service area.

5. Increasing the base station density will increase the capacity of the cellular system? 1 point

- ◉ Yes, if the number of mobile terminals is fixed and their demand is fixed.
- ○ Yes, even if the number of mobile terminals and their demand increase.

6. Select the two most unique features for the next generation cellular system (5G). 1 point

- [x] Ultra-dense base stations
- [x] mmWave transmission
- [ ] Long-distance transmission
- [ ] Frequent handover

7. The uplink power control is a typical example of radio resource management. When in the two-user case, is it optimal to shut off one of the users? 1 point

- ◉ The target SIR is too high.
- ○ One mobile is too close to the base station.

8. In the distributed power control (DPC) algorithm, who measures the SIR (signal to interference), and reports it to whom? 1 point

- ◉ The base station measures it and report to the mobile terminal.
- ○ The mobile terminal measures it and report to the base station.

9. To balance between fairness and throughput-maximization, which objective could be used in packet scheduling? 1 point

- ○ Maximize the minimum throughput of users
- ◉ Proportional fairness

10. What is the most important gain from the packet scheming in the cellular system? 1 point

- ◉ Increasing "average" data rate of each user.
- ○ Save the power consumption of base stations.